SIMATS ENGINEERING

CSA6303- Threat Intelligence and Network Security

A1-Malware Analysis Assignment

Assessment Questions

Subject	Threat Intelligence	**Total Marks**	50
Questions	10 (all compulsory)	**Marks per Question**	5
REg.No	192465008	**Name**	A.TANISHA
Faculty	Dr.V.Parthipan	**Slot**	A

Instructions: Answer all questions. Each question carries 5 marks. Support answers with examples and diagrams where appropriate.

CO3: Perform threat analysis using intelligence models and utilize industry tools for monitoring and detection.

Code of Conduct:

I _S.TANISHA(192465008)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of Student S.TANISHA

Questions :

1. Malware Analysis Using Static Techniques:A security analyst receives a suspicious executable file attached to a phishing email. Before running the file, the analyst wants to examine its properties, file hash, strings, and imported functions. Explain how static malware analysis can help identify suspicious characteristics and assess the potential risk without executing the file.

2. Malware Analysis Using Dynamic Techniques: An organization suspects that an employee has downloaded a malicious program that may modify system files and communicate with an external server. Explain how dynamic malware analysis can be performed in an isolated virtual environment to observe the program's behavior and identify potential Indicators of Compromise (IoCs).

3. Malware Investigation Using a Threat Intelligence Platform: A company's security team discovers a suspicious file on an employee's computer. The file hash appears in a threat intelligence platform associated with a known malware family. Explain how the analyst can use threat intelligence sources to investigate the file, assess the reliability of the information, and recommend suitable containment measures.

4. Malware Detection Through Data Collection and Correlation: A university detects unusual processes on several computers, suspicious DNS requests, and unexpected outbound network connections. Explain how an analyst can collect and correlate endpoint logs, DNS records, firewall logs, and file hashes to determine whether the computers are infected with malware.

5. Malware Investigation Using SIEM Tools:A financial institution's Security Information and Event Management (SIEM) system generates alerts about suspicious process execution, repeated failed logins, and unusual network connections from the same workstation. Describe how the security analyst can use SIEM capabilities to correlate these events, investigate the possible malware infection, and prioritize the incident.

6. Malware Detection and Log Analysis Using Splunk: A company uses Splunk to monitor its network and endpoint logs. The security team suspects that malware is attempting to contact an external command-and-control server. Explain how Splunk can be used to search relevant logs, identify suspicious communication patterns, correlate events, and document evidence of possible malware activity.

7. Malware Investigation Using the ELK Stack: An e-commerce company uses the ELK Stack (Elasticsearch, Logstash, and Kibana) to monitor web server logs, firewall records, and endpoint events. The security team notices repeated connections to suspicious domains and unusual file activity. Explain how the ELK Stack can support malware investigation through log collection, processing, searching, visualization, and correlation.

8. Threat Hunting for Malware Activities: An organization's security team suspects that malware has infected an employee workstation but has not received any antivirus alerts. The workstation shows unusual CPU usage, unfamiliar processes, and connections to unknown external IP addresses. Describe a threat-hunting approach to investigate the workstation, identify suspicious behaviors, collect relevant IoCs, and recommend remediation measures.

9. Comparing Malware Analysis Tools Used in the IT Industry: A cybersecurity team needs to investigate a suspicious file and determine whether it contains malicious code. Compare static analysis tools, dynamic analysis tools, network traffic analyzers, and threat intelligence platforms in terms of their purpose and usefulness. Recommend an appropriate combination of tools for conducting a malware investigation.

10. Malware Analysis Report and Incident Response: A company discovers a suspicious executable on an employee's computer. Initial analysis identifies an unfamiliar process, a modified system configuration, and communication with a suspicious external domain. Prepare an outline for a malware analysis report that includes file information, analysis methodology, observed behavior, IoCs, threat intelligence findings, risk assessment, and recommended containment and recovery actions.

Answers

1. Malware Analysis Using Static Techniques [10 Marks]

Static malware analysis examines a file without executing it, allowing analysts to identify suspicious characteristics safely.

Steps in Static Analysis

1. **Examine File Properties**
    - **Check file name, size, type, creation date, and digital signature.**
    - **Identify whether the file claims to be legitimate software.**
2. **Calculate File Hashes**
    - **Generate MD5, SHA-1, or SHA-256 hashes.**
    - **Compare hashes with malware databases such as VirusTotal.**
3. **Extract Strings**
    - **Analyze readable text within the executable.**
    - **Look for:**
        - **Suspicious URLs**
        - **IP addresses**

        - **File paths**
        - **Registry keys**
        - **Commands**

4. **Analyze Imported Functions**
    - **Identify APIs used by the malware.**
    - **Functions such as:**
        - **CreateProcess()**
        - **WriteFile()**
        - **RegSetValue()**
        - **InternetOpen()**
    - **indicate possible malicious behavior.**
5. **Inspect PE Header**
    - **Check compilation time and entry points.**
    - **Detect packing or obfuscation techniques.**

Benefits

- **Safe analysis without infection risk.**
- **Early identification of malware characteristics.**
- **Collection of Indicators of Compromise (IoCs).**

Conclusion

Static analysis helps security analysts assess a suspicious file's risk by examining its structure, hashes, strings, and imported functions without running the file.

---

2. Malware Analysis Using Dynamic Techniques [10 Marks]

Dynamic analysis involves observing malware behavior during execution in an isolated environment.

Steps

1. Create an Isolated Sandbox

- **Virtual Machine (VM)**
- **Sandbox Environment**
- **No access to production systems**

2. Execute the Program

Monitor the malware while it runs.

3. Observe System Changes

Check for:

- **New files created**
- **Registry modifications**
- **System configuration changes**
- **Added services**

4. Monitor Network Activity

Identify:

- **Outbound connections**
- **DNS queries**
- **Command-and-Control (C2) communication**
- **Data exfiltration attempts**

5. Analyze Processes

Observe:

- **Process creation**
- **Process injection**
- **Privilege escalation**

Indicators of Compromise (IoCs)

- **Malicious domains**
- **Suspicious IP addresses**
- **Registry keys**
- **File hashes**
- **New scheduled tasks**

Conclusion

Dynamic analysis reveals real-time malware behavior and helps identify IoCs that may not be visible during static analysis.

---

3. Malware Investigation Using a Threat Intelligence Platform [10 Marks]

A threat intelligence platform provides information about known threats and malware families.

Investigation Process

Step 1: Search the File Hash

- **Compare the file hash against intelligence databases.**
- **Determine whether it is associated with known malware.**

Step 2: Verify Threat Information

Collect:

- **Malware family name**

- **Threat actor information**
- **Attack techniques**
- **Associated IoCs**

Step 3: Assess Reliability

Validate findings using:

- **Multiple intelligence sources**
- **Vendor reputation**
- **Confidence ratings**
- **Recent threat reports**

Step 4: Determine Impact

Analyze:

- **Affected systems**
- **Potential data loss**
- **Scope of infection**

Containment Measures

1. **Isolate infected endpoints.**
2. **Block malicious IPs and domains.**
3. **Remove malicious files.**
4. **Reset compromised credentials.**
5. **Increase monitoring.**

Conclusion

Threat intelligence helps analysts verify malware, understand risks, and implement effective containment strategies.

4. Malware Detection Through Data Collection and Correlation [10 Marks]

Analysts can identify malware by collecting and correlating information from multiple security sources.

Data Collection Sources

Endpoint Logs

- **Running processes**
- **Application activity**
- **Security events**

DNS Logs

- **Domain requests**
- **Suspicious lookups**

Firewall Logs

- **Inbound and outbound traffic**
- **Blocked connections**

File Hashes

- **Verify suspected files against threat databases**

Correlation Process

Source	Information
Endpoint Logs	Suspicious process execution
DNS Logs	Requests to malicious domains
Firewall Logs	Connections to suspicious IPs

File Hashes	**Known malware signatures**

Analysis

- **Identify systems showing all indicators.**
- **Determine infection timeline.**
- **Identify malware communication channels.**

Conclusion

Correlating logs from multiple sources provides stronger evidence of malware infection than relying on a single data source.

---

5. Malware Investigation Using SIEM Tools [10 Marks]

A SIEM helps analysts detect malware by aggregating and correlating security events.

Investigation Steps

1. Collect Logs

Gather data from:

- **Endpoints**
- **Firewalls**
- **Servers**
- **Authentication systems**

2. Correlate Events

Example:

Event	Observation
Failed Logins	Multiple attempts
Suspicious Process	Unknown executable
Network Activity	Unusual external connection

The combination may indicate malware infection.

3. Timeline Analysis

Determine:

- **Initial infection**
- **User activity**
- **Malware execution**
- **Network communication**

4. Threat Prioritization

Assess:

- **Severity**
- **Criticality of affected asset**
- **Business impact**

Benefits

- **Centralized visibility**
- **Faster investigation**
- **Improved incident response**

Conclusion

SIEM tools help correlate suspicious events, identify malware activity, and prioritize incidents based on risk.

---

6. Malware Detection and Log Analysis Using Splunk [10 Marks]

Splunk enables real-time monitoring and analysis of malware-related activities.

Investigation Process

Search Endpoint Logs

Identify:

- **Unknown processes**
- **Application crashes**
- **Unauthorized software execution**

Analyze Network Logs

Search for:

- **Connections to external IPs**
- **Repeated DNS lookups**
- **Beaconing patterns**

Correlate Events

Example:

1. **Suspicious process starts.**
2. **DNS request to unknown domain.**
3. **Outbound traffic to external server.**

Together these indicate possible malware activity.

Create Dashboards

Visualize:

- **Infected hosts**
- **Alert frequency**
- **Communication patterns**

Preserve Evidence

Document:

- **Timestamps**
- **Log entries**
- **IP addresses**
- **File hashes**

Conclusion

Splunk helps analysts search logs, correlate events, visualize suspicious activities, and document evidence for malware investigations.

---

7. Malware Investigation Using the ELK Stack [10 Marks]

The ELK Stack consists of:

- **Elasticsearch (Storage and Search)**
- **Logstash (Log Processing)**
- **Kibana (Visualization)**

Investigation Process

Log Collection

Logstash collects:

- **Web server logs**

- **Firewall logs**
- **Endpoint logs**

Processing

Logstash:

- **Parses data**
- **Removes inconsistencies**
- **Normalizes fields**

Searching

Elasticsearch enables:

- **Fast searches**
- **IOC queries**
- **Timeline analysis**

Visualization

Kibana provides:

- **Dashboards**
- **Graphs**
- **Malware activity trends**

Correlation

Analysts correlate:

- **Suspicious domains**
- **File modifications**
- **Endpoint activity**
- **Network connections**

Conclusion

The ELK Stack helps security teams collect, process, search, visualize, and correlate logs for effective malware investigations.

---

8. Threat Hunting for Malware Activities [10 Marks]

When antivirus does not generate alerts, threat hunting is used to proactively search for hidden malware.

Investigation Steps

1. Analyze System Performance

Check:

- **CPU usage**
- **Memory usage**
- **Disk activity**

2. Investigate Running Processes

Identify:

- **Unknown executables**
- **Unsigned applications**
- **Suspicious services**

3. Examine Network Connections

Look for:

- **External IP communications**
- **Unknown domains**
- **Frequent beaconing**

4. Search for Persistence Mechanisms

Check:

- **Startup folders**
- **Scheduled tasks**
- **Registry run keys**

5. Collect IoCs

Gather:

- **File hashes**
- **Domains**
- **IP addresses**
- **Process names**

6. Verify Using Threat Intelligence

Compare observed IoCs with threat intelligence databases.

Remediation

- **Isolate workstation.**
- **Remove malware.**
- **Change passwords.**
- **Patch vulnerabilities.**
- **Deploy EDR monitoring.**

Conclusion

Threat hunting identifies hidden malware through behavioral analysis, IoC collection, and proactive investigation techniques.

---

9. Comparing Malware Analysis Tools Used in the IT Industry [10 Marks]

Tool Category	Purpose	Advantages
Static Analysis Tools	Examine files without execution	Safe and quick
Dynamic Analysis Tools	Monitor runtime behavior	Reveals actual actions
Network Traffic Analyzers	Inspect network communications	Detects C2 traffic
Threat Intelligence Platforms	Verify known threats	Provides threat context

Examples

Static Analysis Tools

- **PEStudio**
- **Strings**
- **ExifTool**

Dynamic Analysis Tools

- **Cuckoo Sandbox**
- **Any.Run**
- **VMware Sandbox**

Network Traffic Analyzers

- **Wireshark**
- **Zeek**
- **TCPDump**

Threat Intelligence Platforms

- **VirusTotal**
- **AlienVault OTX**
- **MISP**

Recommended Combination

1. **Static Analysis → Initial inspection.**
2. **Dynamic Analysis → Behavior analysis.**
3. **Network Analysis → Communication monitoring.**
4. **Threat Intelligence → Threat verification.**

Conclusion

Combining all four tool categories provides comprehensive malware investigation and accurate threat assessment.

---

10. Malware Analysis Report and Incident Response [10 Marks]

Malware Analysis Report

1. Executive Summary

A suspicious executable was discovered on an employee workstation and exhibited potentially malicious behavior.

2. File Information

- **File Name**
- **File Size**
- **File Type**
- **MD5/SHA256 Hash**
- **Detection Date**

3. Analysis Methodology

Static Analysis

- **Hash analysis**
- **String analysis**
- **Import analysis**

Dynamic Analysis

- **Sandbox execution**
- **Behavior monitoring**

4. Observed Behavior

- **Unknown process execution**
- **Registry modifications**
- **System configuration changes**
- **External network communication**

5. Indicators of Compromise (IoCs)

Category	Indicator
File	SHA256 Hash
Process	Suspicious Executable
Network	Malicious IP Address
Domain	Suspicious Domain
Registry	Persistence Key

6. Threat Intelligence Findings

- **Malware family identified.**

- **Associated attacker techniques.**
- **Known indicators matched.**

7. Risk Assessment

Factor	Assessment
Severity	High
Likelihood	High
Business Impact	Significant

8. Containment Actions

- **Isolated infected host.**
- **Blocked malicious domains.**
- **Disabled compromised accounts.**

9. Recovery Actions

- **Malware removal.**
- **Password reset.**
- **System restoration.**
- **Security patch deployment.**

10. Recommendations

1. **Enable EDR solutions.**
2. **Enforce Multi-Factor Authentication (MFA).**
3. **Improve email filtering.**
4. **Conduct regular threat hunting.**
5. **Provide security awareness training.**

Conclusion

The malware analysis report documents the investigation process, identified IoCs, observed behavior, risk assessment, and remediation actions required to contain the threat and restore secure operations.